

Zhiheng Zhang

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github.com/zhiheng-zhang-Mera

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Research Profile

Systems research engineer interested in bounded edge execution, trustworthy task continuity and long-horizon personal agents under resource constraints.

Education

University of Melbourne, Australia	2025–2026 (expected)
Master of Computer Science (research project underway)	
University of British Columbia, Canada	Completed 2024
Bachelor of Science in Computer Science	

Selected Research Engineering Projects

Utopia / Digital City (Completed, Frozen)	Integrated successor
github.com/zhiheng-zhang-Mera/utopia	

- Completed and froze Utopia as the integrated successor to DS-Hns and Codex Boss; delivered Web/Android control, authenticated routing and persistent task-state contracts.
- Verified a three-end setup with two physical Windows execution workers and Android control; PCF integration includes bounded real cross-host task results and explicit resource-observation contracts.
- Maintained research-evaluation evidence for fault injection, replay, traceable results and independent bounded reproducibility; wearable deployment and full dual-server failover remain future work.

DS-Hns (Completed, Frozen)	Predecessor: durable agent execution
github.com/zhiheng-zhang-Mera/DS-Hns	

- Completed and froze predecessor DS-Hns with durable repository-work lifecycles, scheduling, restart/resume recovery and isolated extensions.
- Used explicit terminal-state and recovery evidence to study the gap between continued process execution and genuine task completion.

Codex Boss (Completed, Frozen)	Predecessor: agent governance
github.com/zhiheng-zhang-Mera/Codex-Boss	

- Completed and froze predecessor Codex Boss with agent orchestration, critique and evidence-governed acceptance foundations.
- Preserved unresolved negative cases and traceable failures rather than treating superficial success signals as proof of completion.

Research Methods and Skills

System prototyping; multi-device software integration; long-running task lifecycles; fault injection; reproducible experiment and evidence design; test and failure analysis; independent cross-host review.

Proposed Doctoral Research

How a personal hub can execute authorized agent tasks reliably while keeping lightweight wearable clients responsive and avoiding unnecessary migration to other devices. Experimental priority: Compare goal-aware admission, recovery and completion receipts against naive offload policies; evaluate task outcome, failure handling, unintended user interruption, locality and latency on a bounded edge testbed.

Future project: Celestial is the planned redesign and next iteration of the completed, frozen Utopia; it is not yet a completed product.